



WL - MHT - 4G

4G Mobile Handset Development Lab

EXPERIMENTAL MANUAL

.... A MESSAGE FROM

Today's system designers are faced with tomorrow's problems. Digital Communication is one of the important subject need to teach while learning electronics.

It is our vision to provide with the product you need for training ensuring lasting reliability & quality.

OUR MOTTO;

- Light years ahead – refers to leadership.

As leaders in our industry in India, We are totally committed to servicing as the standard against which all are measured in the areas of

• Design • Quality • Value • Delivery • Support.

We are truly light years ahead of our competition in this area. That means that you our valued customers are guaranteed satisfaction.

As you will move through this manual you will quickly discover that We have a complete, truly innovative & superior training products we are so committed to quality that we back our products with a complete comprehensive warranty.

AKADEMIKA LAB SOLUTIONS

Sr.No.15/8/1,Unit No. 9 Kruti Industrial Estate,
Opp Karishma Society, Off Karve Road,Kothrud,

Pune – 411038,India

ContactNo.: +91-7447438443

Email: lab@akademika.in

www.akademika.in

SAFETY RULES

- Read carefully and follow the instructions mentioned in this manual. This user manual includes all the important points about the installation, use and the maintenance of the product. Keep this manual always with you, for quick reference.
- After unpacking the product, arrange all the accessories in proper order, so that their integrity is checked with the packing list. Also, ensure that the accessories have no visible damage.
- Before connecting the power supply to the kit, be sure that the jumpers and the connecting chords are connected correctly, as per the experiment.
- This kit must be employed only for the use for which it has been conceived, i.e. as educational kit and must be used under the direct survey of expert personnel. Any other use is inadvisable and dangerous too. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.
- In case of any fault or malfunctioning in the trainer kit, turn off the power supply. Please do not tamper the kit. If you require our service, kindly contact the service centre for technical assistance.
- The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

WARRANTY

This kit is warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we would charge for repairs, regardless of the warranty period. The warranty does not cover include perishable items like connecting chords, crystals, etc. and other imported items.

Conditions and Limitations

The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product will be Akademika Lab Solutions sole judgment. The defective product will be replaced with a new one or repaired, without charge or with charge.

In the warranty period if the service is needed, the purchaser should get in touch with the service center or the sales outlet. The purchaser should return the product to the service center or the sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of the kit has to be mentioned specifically. We return the product to the purchaser at our expense. In case the warranty does not cover the product on Akademika Lab Solutions judgment, we would repair the product after obtaining prior permission from purchaser who would receive an estimate statement about the repairing charges. In such cases, Akademika Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but they have no rights to stretch the meaning of original warranty contents or to offer an additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs, maintenance and up gradation of products, be sure to contact our service center or the sales outlet.

SR NO:	INDEX	PAGE NO:
I	TECHNICAL SPECIFICATIONS	9
II	INTRODUCTION	13
III	PACKING LIST	32
IV	SOFTWARE INSTALLATION	
	1. INSTALLATION OF JAVA IDE ENVIRONMENT	34
	2. INSTALLATION OF ANDROID STUDIO	40
	3. INSTALLATION OF ADB DRIVER	45
	4. INSTALLATION OF 4G MOBILE TRAINER SOFTWARE	47
V	WL - MHT - 4G EXPERIMENTS DETAILS	
	1. STUDY OF ANDROID MOBILE APPLICATIONS IN PLAYSTORE	50
	2. STUDY OF ANDROID STUDIO PROGRAMS	59

TECHNICAL FEATURES & SPECIFICATIONS

FEATURES:-

- Support worldwide LTE-FDD, LTE-TDD, DC-HSDPA, DC- HSUPA, HSPA+, and HSDPA, HSUPA, WCDMA, TD-SCDMA, EVDO/CDMA, EDGE and GPRS coverage.
- Support short-range wireless communication via Wi-Fi 802.11a/b/g/n/ac and BT4.2 LE standards.
- Integrate GPS/GLONASS/BeiDou satellite positioning systems.
- Support multiple audio and video codecs
- Built-in high performance Adreno™ 506 graphics processing unit.
- Provide multiple audio and video input/output interfaces as well as abundant GPIO interfaces.
- Supports Text and PDU mode, Point - to point MO and MT, SMS cell broadcast.
- Supports G711, QCELP, EVRC, EVRC-B, EVRC-WB, AMR-NB, AMR-WB, GSM-EFR, GSM-FR, GSM-HR Audio Codec.
- Based on Android OS 7.1
- 5 groups of I2C, used for peripherals such as TP, camera, sensor etc.
- Support for I2S Peripherals.
- 2 general purpose ADC interfaces supports up to 15-bit sampling accuracy.
- Supports real time clock.

SPECIFICATION: -

Module: LTE Module

- Hexagon DSP v56 core up to 850MHz, 768KB L2 cache, MEMORY 16GB eMMC + 2GB LPDDR3(default), 32GB eMMC + 4GB LPDDR 3 (optional), Operating System Android OS 7.1 Modem interface with GUI and ARM processor.

Power Supply

- DC power supply: 4.75V~6.0V (typical: 5.0V)
- VBAT : 3.55V~4.4V (typical: 3.8V)

LCM Interfaces

- Two 5-inch 1280 × 720 HD resolution LCD

Touch Panel Interfaces

- One capacitive touch panel

Camera Interfaces

- Rear camera supports 16M pixels
- Front camera supports 8M pixels

USB Interface

- USB type-C interface, Compliant with USB 3.0 and 2.0 specifications, with transmission rates up to 5Gbps on USB 3.0 and 480Mbps on USB 2.0, Support USB OTG, used for data transmission, software debugging and firmware upgrade.

Audio Interfaces

Analog Input: 3 groups of analog microphone inputs, integrating internal bias voltage

- One headset input
- One ECM-type microphone input
- One MEMS-type microphone input

Analog Output: 3 groups of analog audio outputs, including

- One stereo headset output
- One mono earphone differential output
- One mono loudspeaker differential amplifier output

(U)SIM Interfaces

- Support USIM/SIM cards: 1.8V and 2.95V, Support dual SIM dual standby

UART Interfaces

- Main UART for data transmission and AT command communication
Debug UART for debugging

SD Card Interface

- Support 4-bit SD card with hot-plug detection, MIPI+SPI/I2S Interface
Reserved

Antenna Interfaces

- main antenna
- Rx-diversity/MIMO antenna
- GNSS antenna
- Wi-Fi/BT antenna

Flashlights

- Support 2 flashlight LEDs for testing the module's flashlight interfaces

Sensors

- Support three sensors including ALS/PS, accelerometer/gyroscope and compass

Vibrator

- A vibrator for testing the motor drive interface of the module

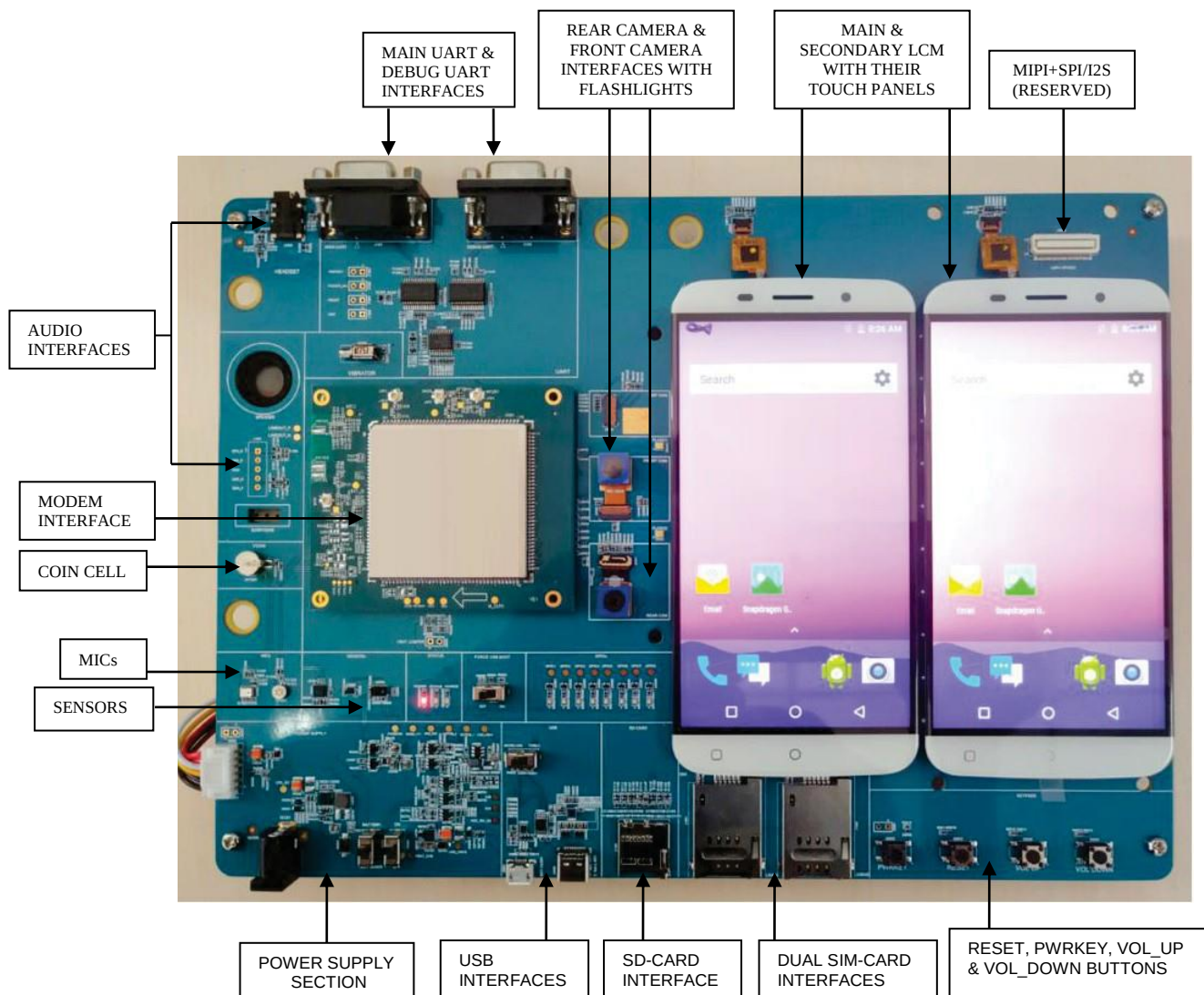
Switches and Buttons

- 4 switches including two power supply switches, one emergency, download switch and one USB switch; 4 buttons including RESET (reserved), PWRKEY, VOL_UP and VOL_DOWN;

Status Indication LEDs

- 3 LEDs are available for signal indication

INTRODUCTION



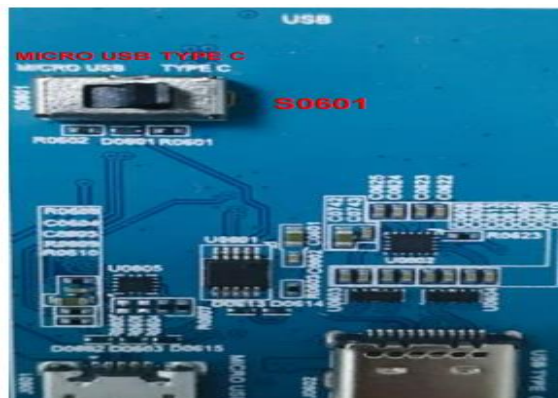
Interfaces of **WL - MHT - 4G** Trainer Board:

- **Power Supply Interface:**

Power Supply	J0201	The power jack on the Board. Typical power supply: +5.0V Li-polymer battery connector
	J0202	

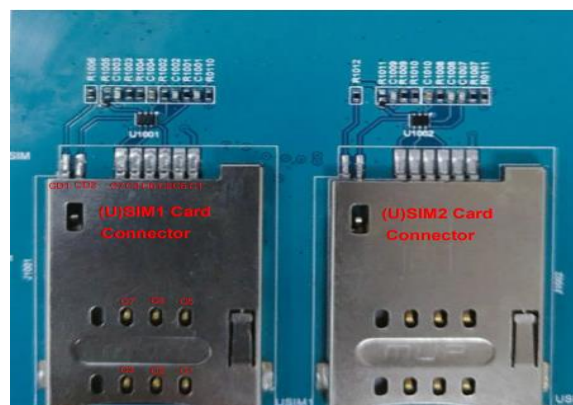
The WL - MHT - 4G Trainer Board is powered by an external 5.0V DC power adapter. Connect the DC power adapter to the power jack (J0201)

The following figure shows the USB interfaces and S0601 switch.

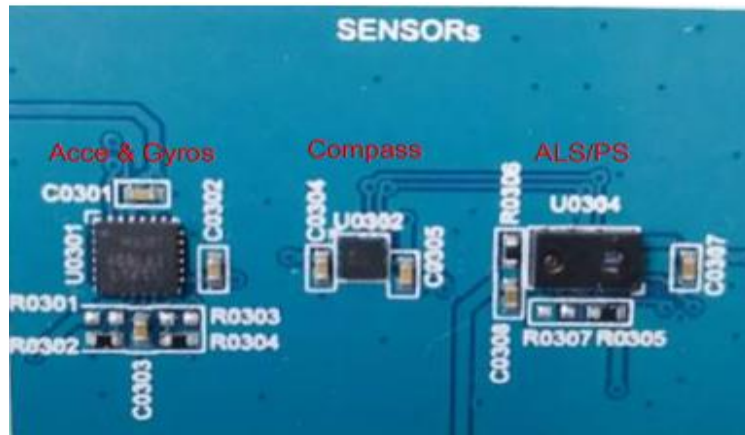


- Audio Interfaces:
 - Speaker
Used for loudspeaker
Rated resistance: 8Ω
Rated power: 1.5W
 - Earphone
Used for earphone
Rated resistance: 32Ω
Rated power: 30mW
 - MIC (MEMS)
Used for MEMS-type microphone
 - MIC (ECM)
Used for ECM-type microphone
 - J1202
Audio Jack for headset
- (U)SIM Interfaces:

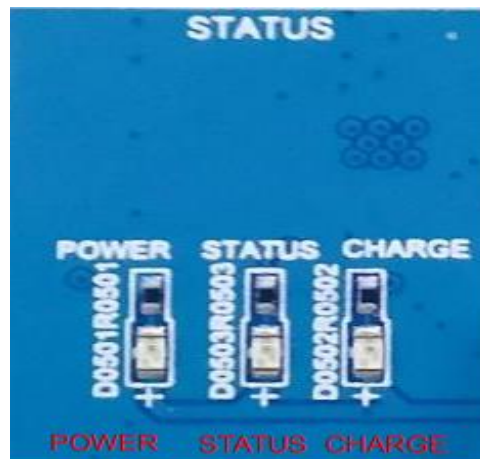
The WL - MHT - 4G Trainer provides two 6-pin push-push type (U) SIM card (3V or 1.8V) connectors J1001 ((U) SIM1 card connector) and J1002 ((U) SIM2 card connector)



- Sensors:**
 The WL - MHT - 4G Trainer provides three sensors for testing purpose, as shown in the figure below.



- Status Indication LEDs**
 There are several status indication lights (D0501, D0502 and D0503) for signal indication on WL - MHT - 4G Trainer, shown as below.



D0501	POWER	Indicate the power supply status of Smart modules Bright: POWER ON Extinct: POWER OFF
D0502	STATUS	Indicate the operation status of Smart modules
D0503	CHARGE	Indicate the charge status of Smart modules Bright: module is charging Extinct: module charge is stopped

OPERATION PROCEDURES ILLUSTRATION

- Power ON Smart Modules:

The following are the procedures for powering on the smart modules.

1. The WL - MHT - 4G Trainer Board is powered by an external 5.0V DC Supply, so connect the DC power adapter to the power jack (J0201) and make mode of switch S0201 to DC POWER state.
2. Switch S0202 to ON state, and then D0501 will be lighted.
3. Now, connect the C-Type usb cable to PC and make mode of switch S0601 to Type-C mode.
4. Also connect Wi-Fi/BT antenna, GNSS antenna and main antennas with RF cables to the appropriate connectors on module board.
5. Press D0501 (PWRKEY) for at least 2s to turn on the module. When the following interface is displayed on the utilized LCD(s), it means the booting has been completed.

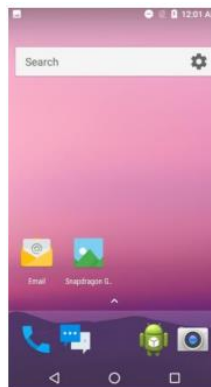


Figure: LCD Display Indicating Module's Power-on

- Power OFF Smart Modules

There are two methods to power off the module.

The steps for the first method are as follows:

1. Press D0501 (PWRKEY) for at least 1s under the power-on state of the module, and then LCD will display a menu for selection shown as the following figure:



Figure: LCD Menu Display for Powering off Module

2. Choose "Power off" option.
 3. The module will be powered off normally.
- The other method is to press down PWRKEY for at least 8s until the module is shut down completely

PACKING LIST

Table: Accessories List

Items	Description	Quantity
Cables	USB Type-C cable	1
	RF cables	4
	USB to UART converter cable	1
Antennas	Main antennas	2
	Wi-Fi/BT antenna	1
	GNSS antenna (passive)	1
Adapter	5.0V DC power supply	1
Others	Bolts and coupling nuts for fixing trainer board	4 for each
Pendrive	Software Installation Data pendrive.	1
audio	Earphone	1



SOFTWARE INSTALLATION

1) System requirements:

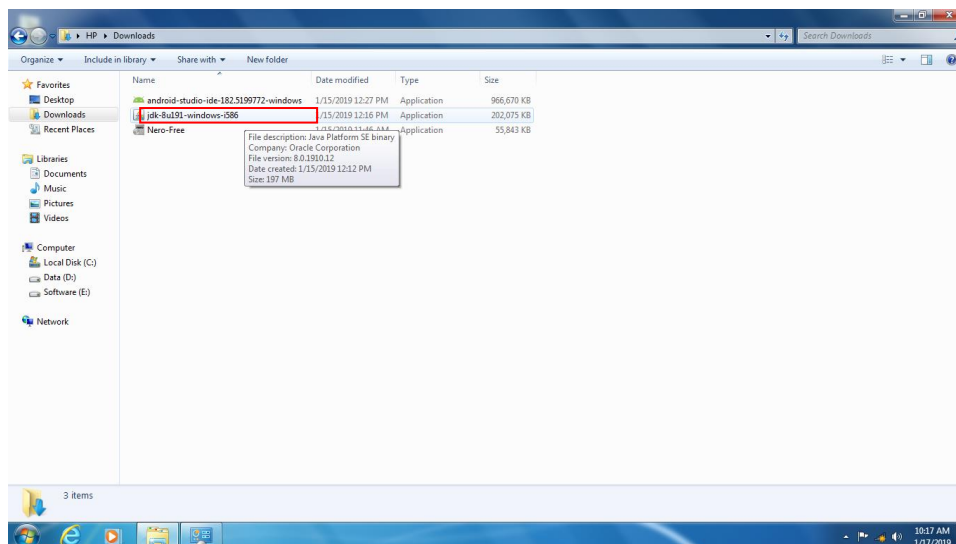
- **Operating system requirements.**
Windows 7 / Windows 10
- **Environment requirement.**
JAVA IDE
- **Android Studio software for android development.**

2) Hardware System Requirements:

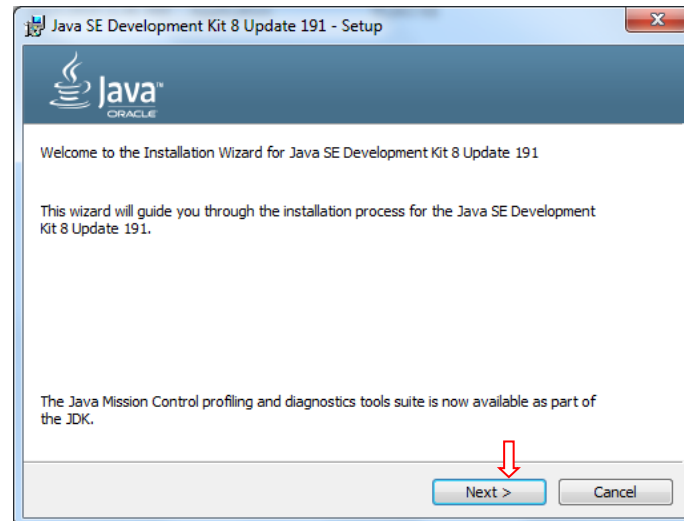
- **CPU**
Intel 5405U 2.3G
- **Minimum RAM**
4GB
- **USB Ports :**
One number of USB 2.0 port.

I. INSTALLATION OF JAVA IDE ENVIRONMENT

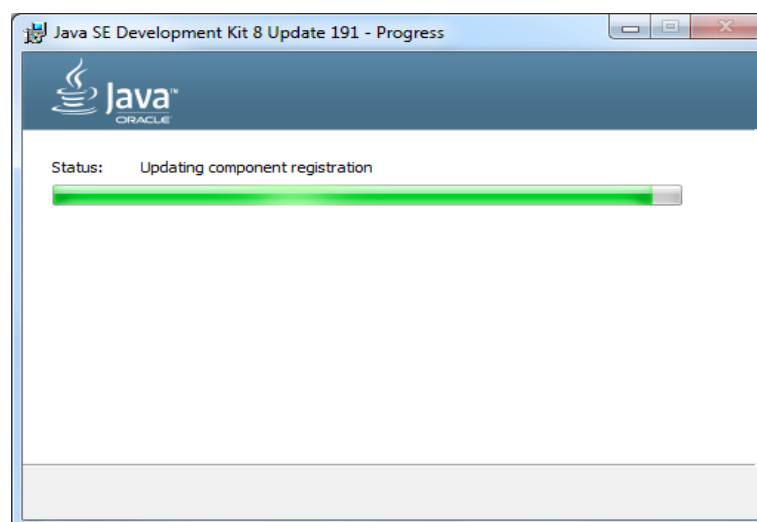
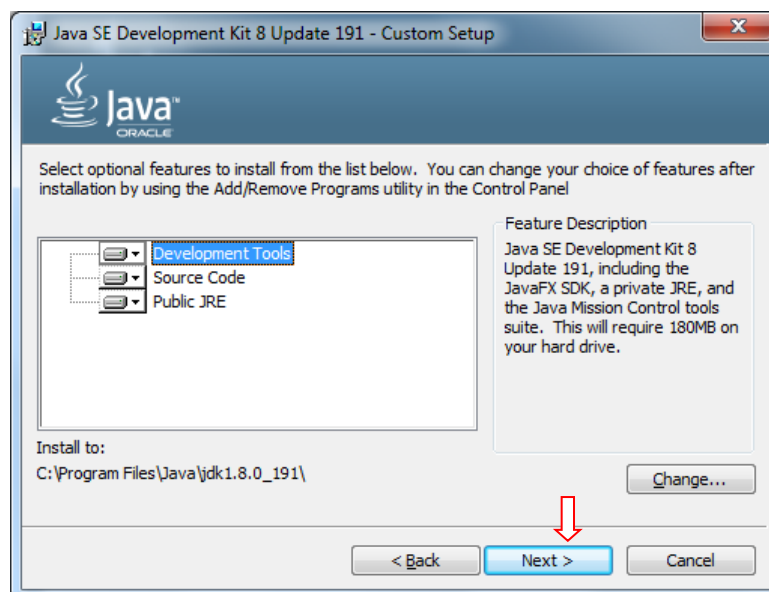
I. Double Click on jdk-8u191-windows-i586.



II. Click on Next.



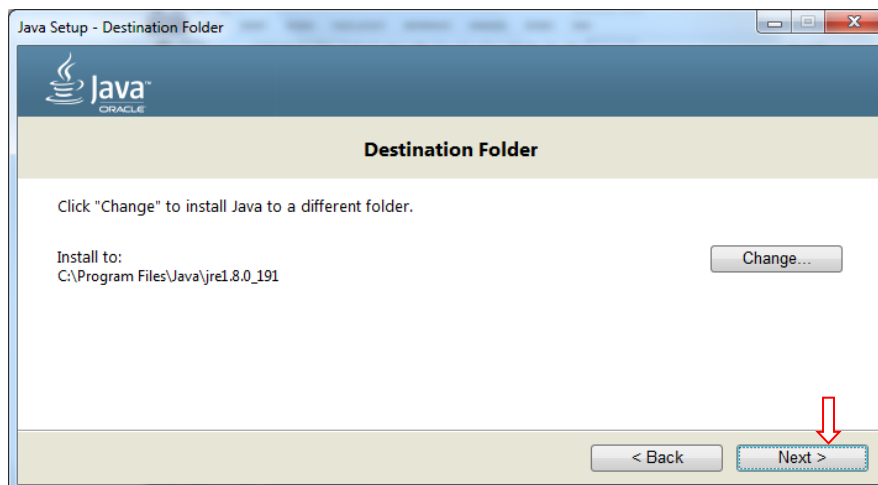
III. Click next.



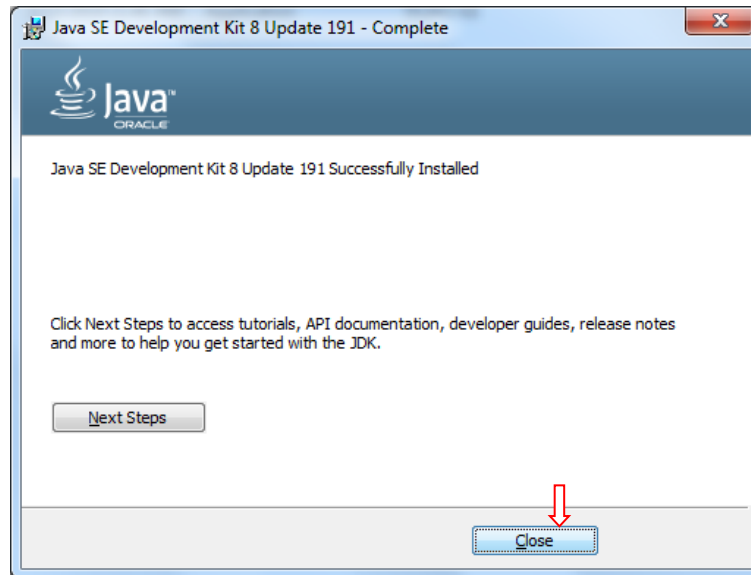
IV. Click on OK.



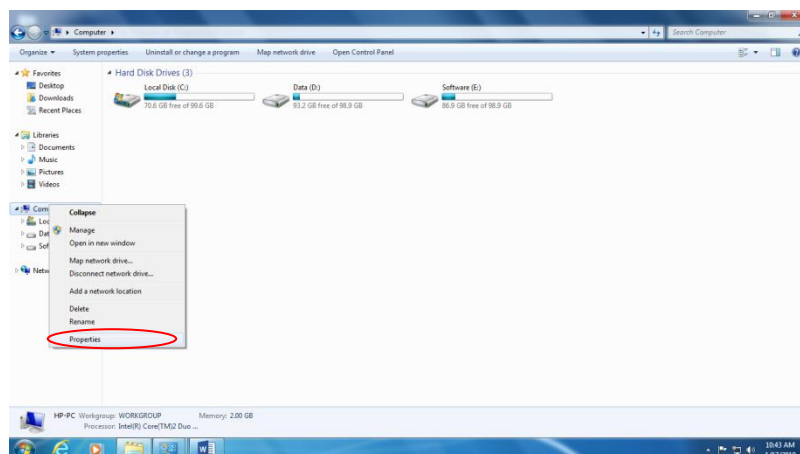
V. Click Next.



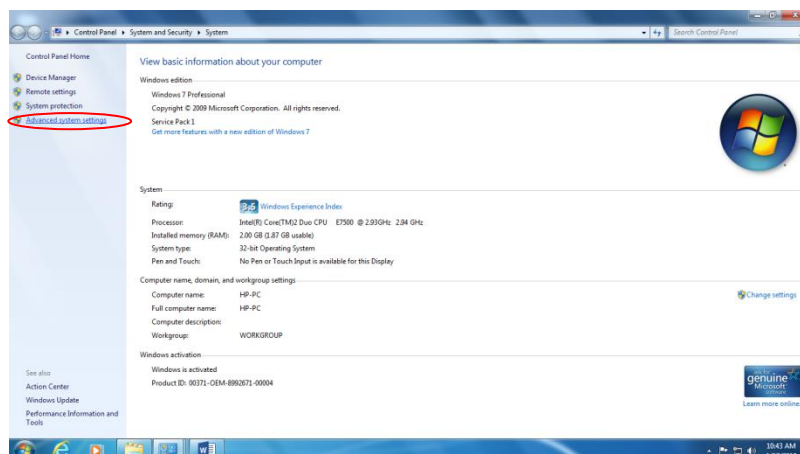
VI. Installation is completed, now click Close.



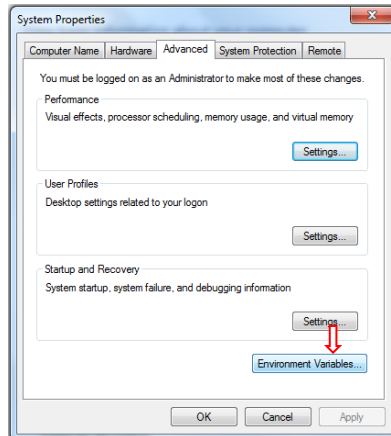
VII. To set environment, go to My Computer Properties.



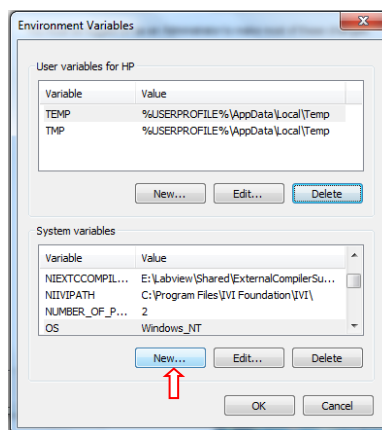
VIII. Now click on Advanced System Settings.



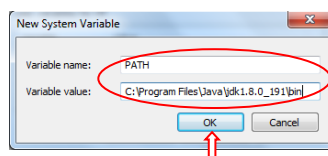
IX. Click on Environment Variables



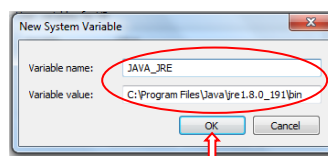
X. Click on New to create new variable



XI. Now Type PATH in variable name and directory path in variable value and Click Ok.



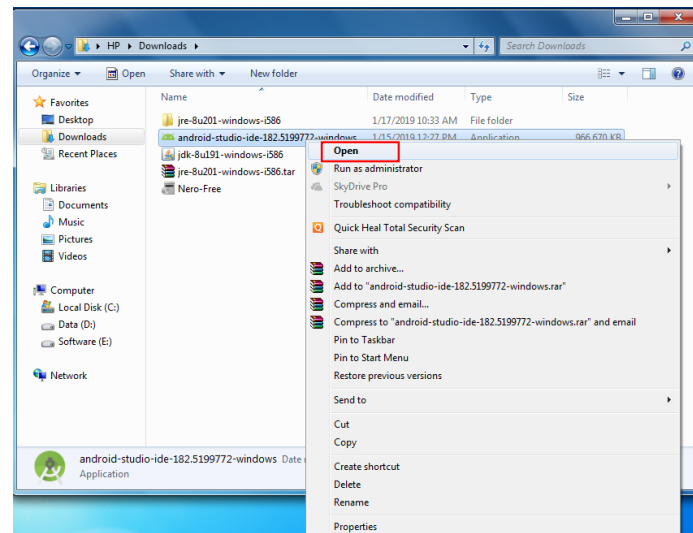
XII. Now Type JAVA_JRE in variable name and directory path in variable value and Click Ok.



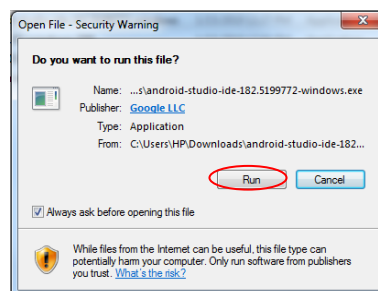
II. INSTALLATION OF ANDROID STUDIO

Note: The Computer Should Be Connected to Internet While Installation Procedure of Android Studio to get all updated library files.

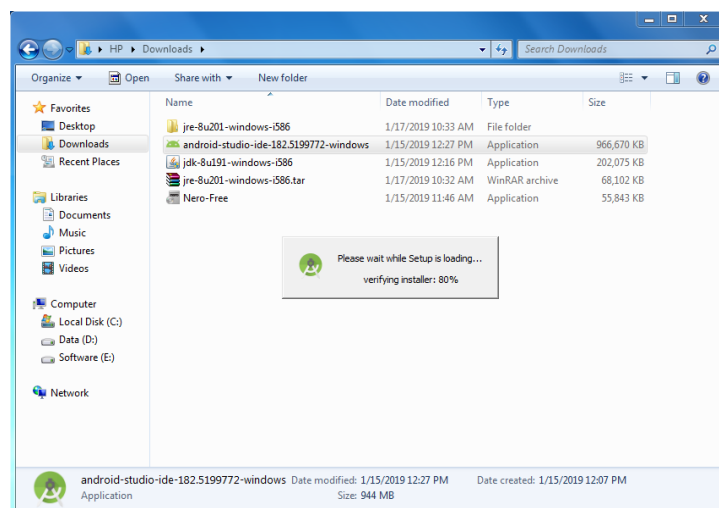
I. Open android-studio-ide-182_5199772-windows.



II. Click on Run.



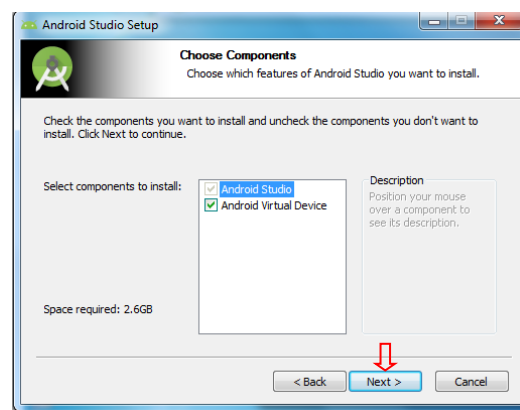
III. Wait till setup is loading.



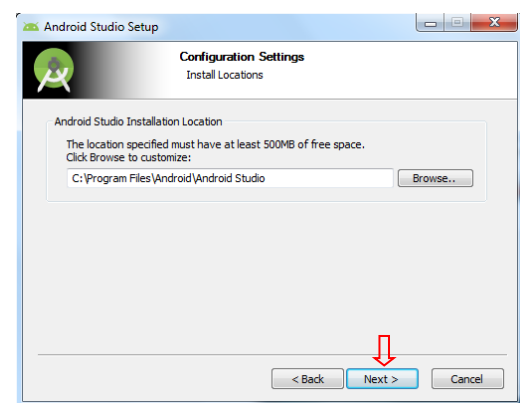
IV. Click Next.



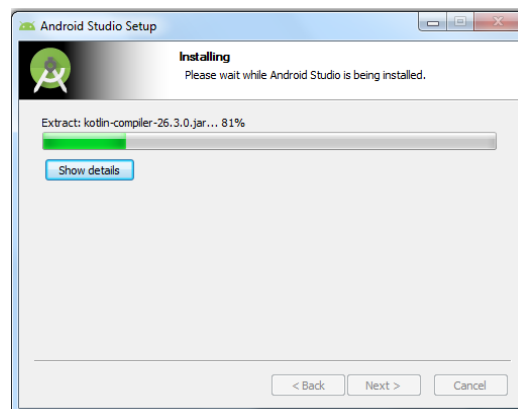
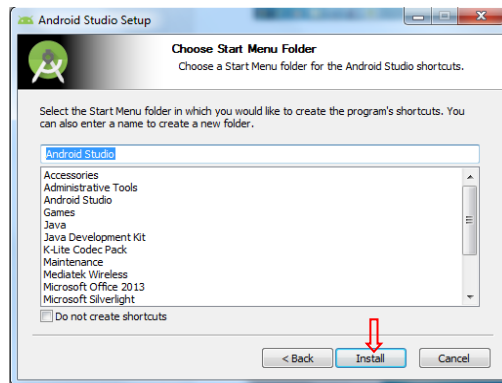
V. Click Next.



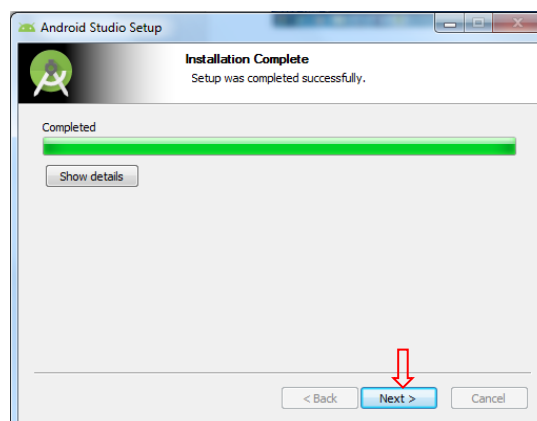
VI. Click Next.



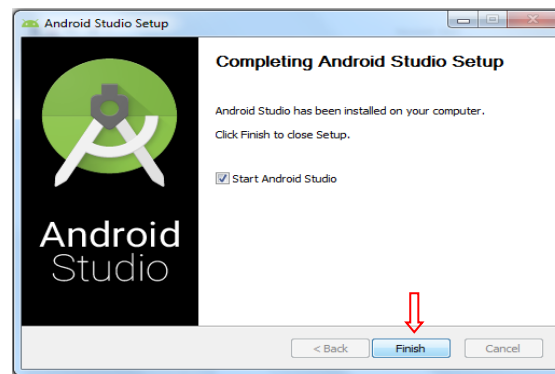
VII. Click Install.



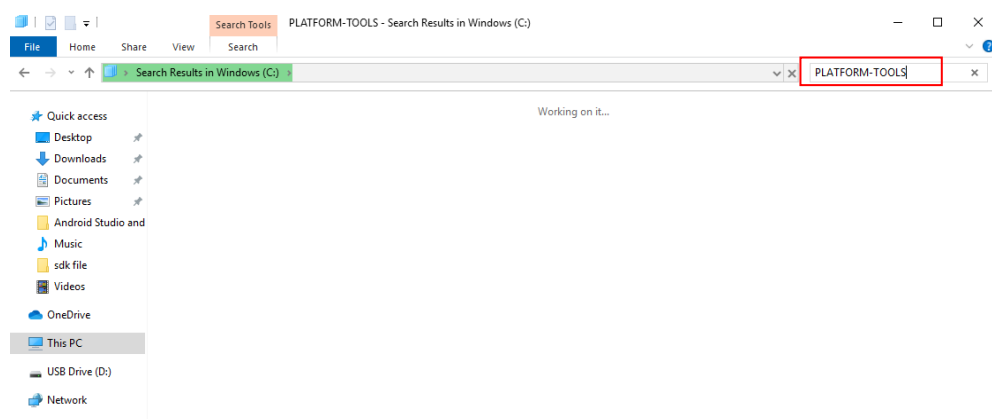
VIII. Click Next.



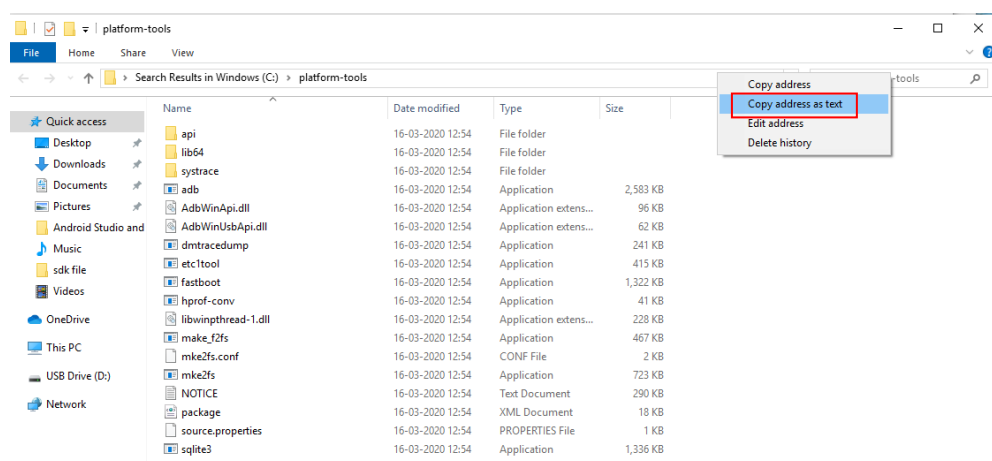
IX. Click Finish.



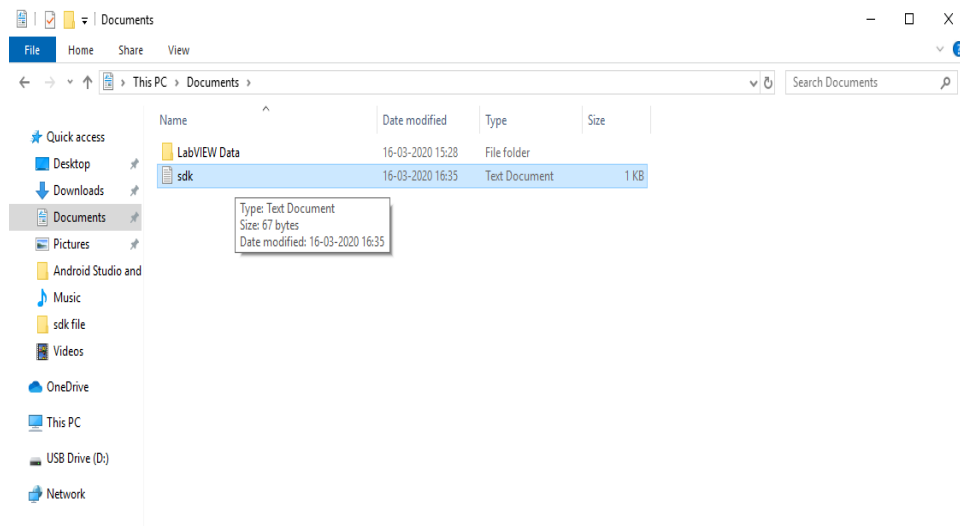
X. Now, go to C drive and search the term '**platform-tools**' in search option.



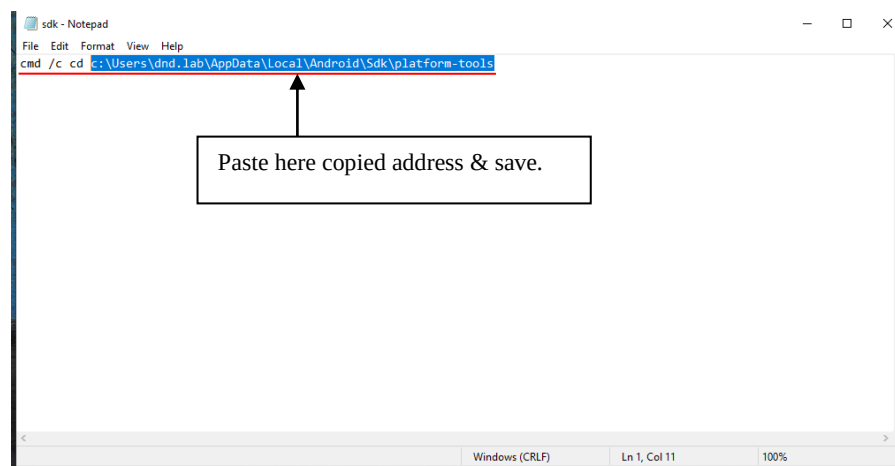
XI. Open the listed '**platform-tools**' folder and Right click on address bar and select '**copy address as text**' option.



XII. Make a new text (.txt) file and name it as 'sdk' in document directory.

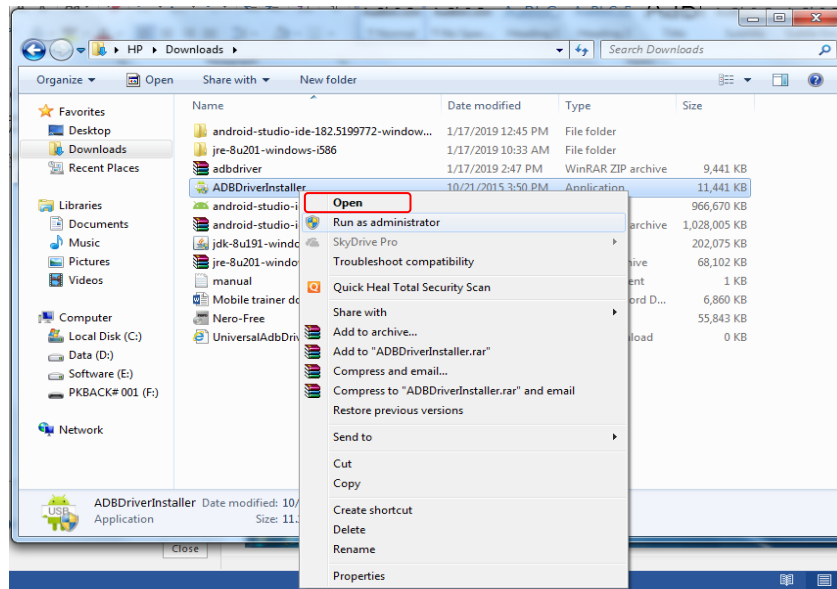


XIII. Write following text in sdk.txt file and save it to document directory.

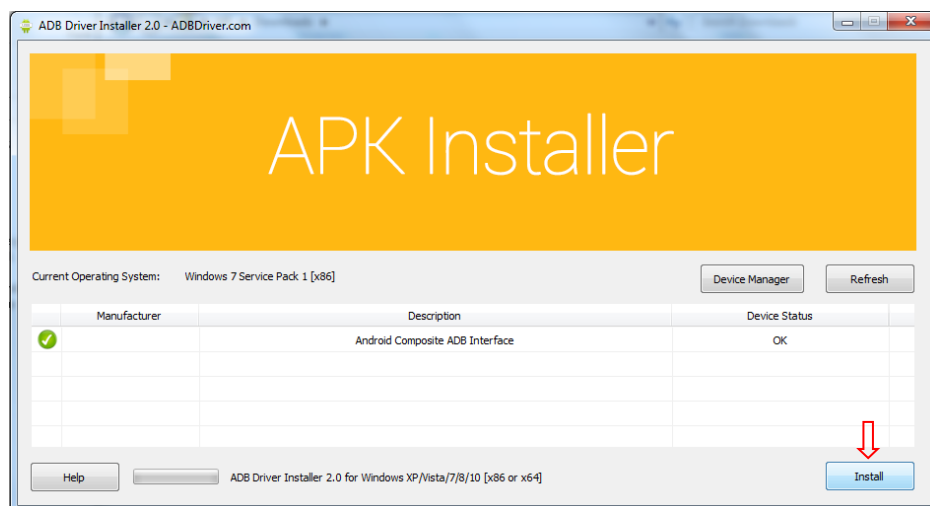


III. INSTALLATION OF ADB DRIVER

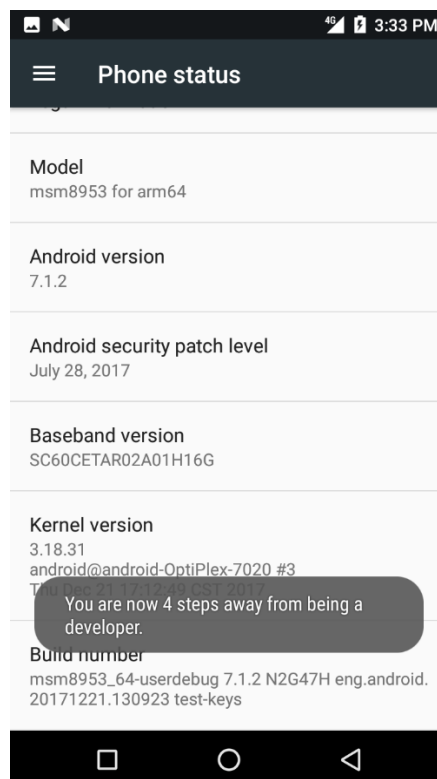
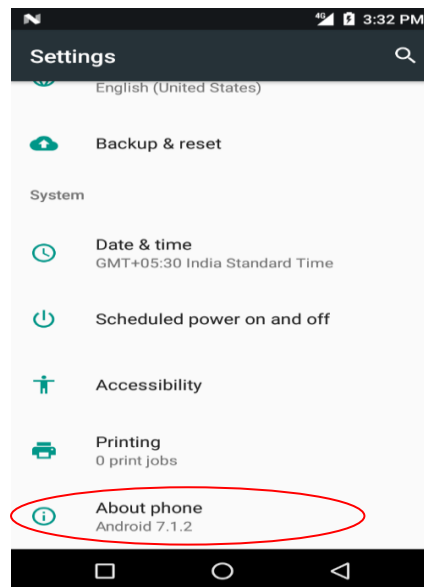
1. Connect C-Type usb connector to 4G Mobile Trainer Kit and Computer.
2. Now Right click on ADB Driver Installer and click on Run as administrator.



3. Click Install

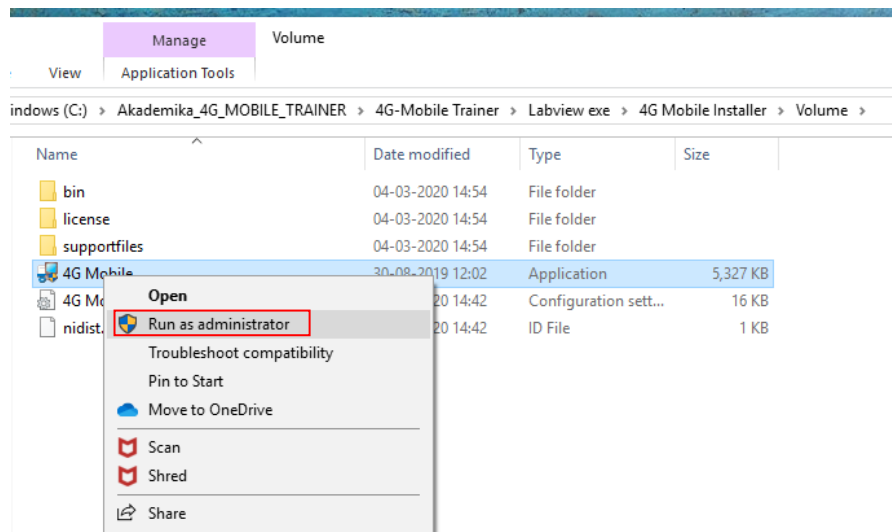


4. Check a Phone status in mobile settings option as per following Screenshots.

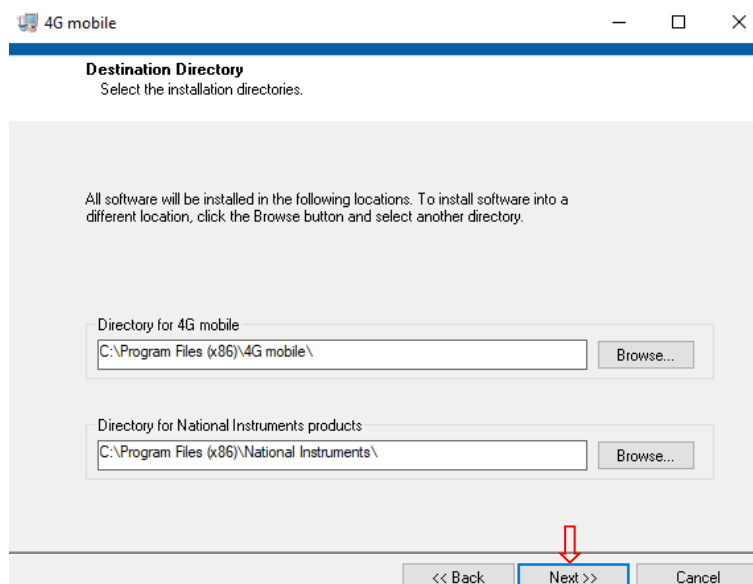


IV. INSTALLATION OF 4G MOBILE TRAINER SOFTWARE

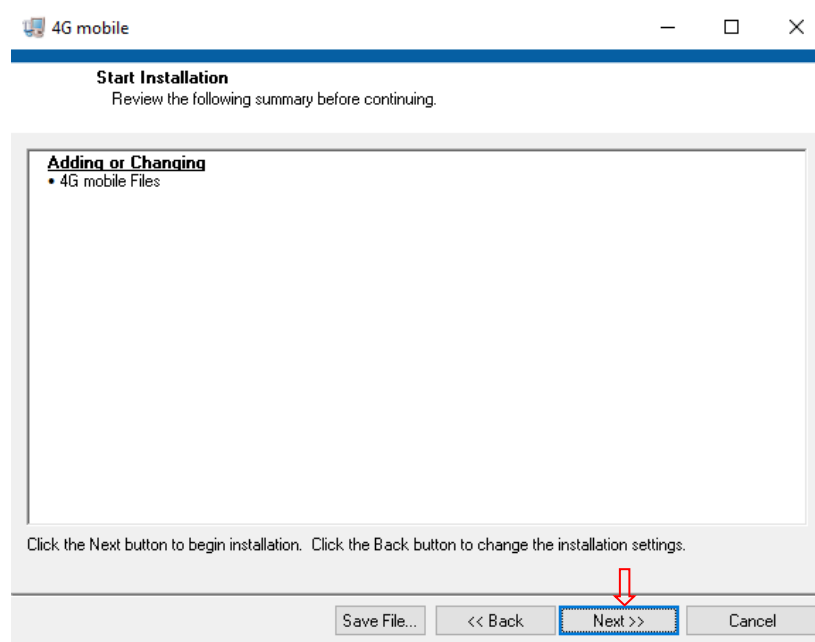
I. Now, install 4G mobile application.



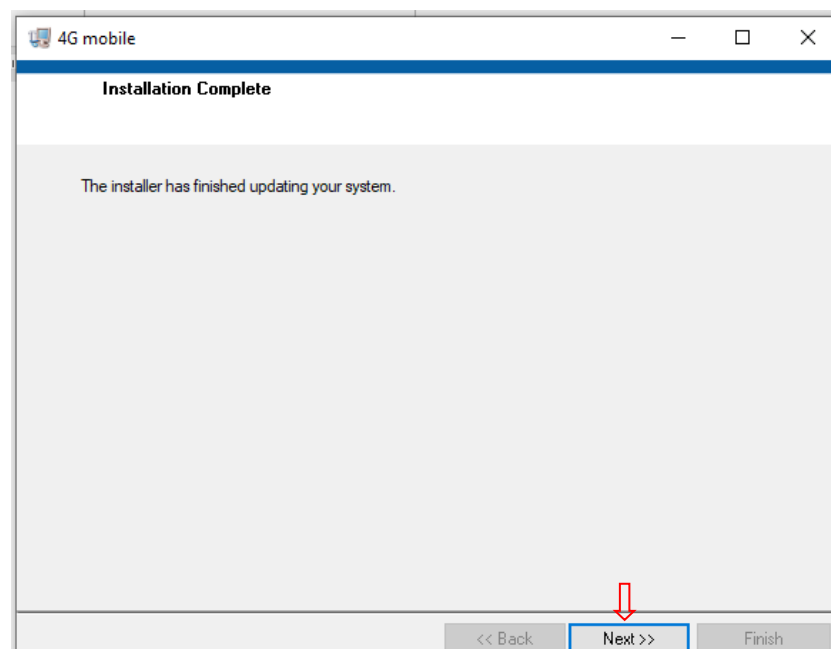
II. Click on next.



III. Again, click on next



IV. Click Next.



V. Following software icon of 4G Mobile will be generated on desktop after successful installation.



EXPERIMENT

STUDY OF ANDROID MOBILE APPLICATIONS IN PLAYSTORE

STUDY OF ANDROID MOBILE APPLICATIONS IN PLAYSTORE

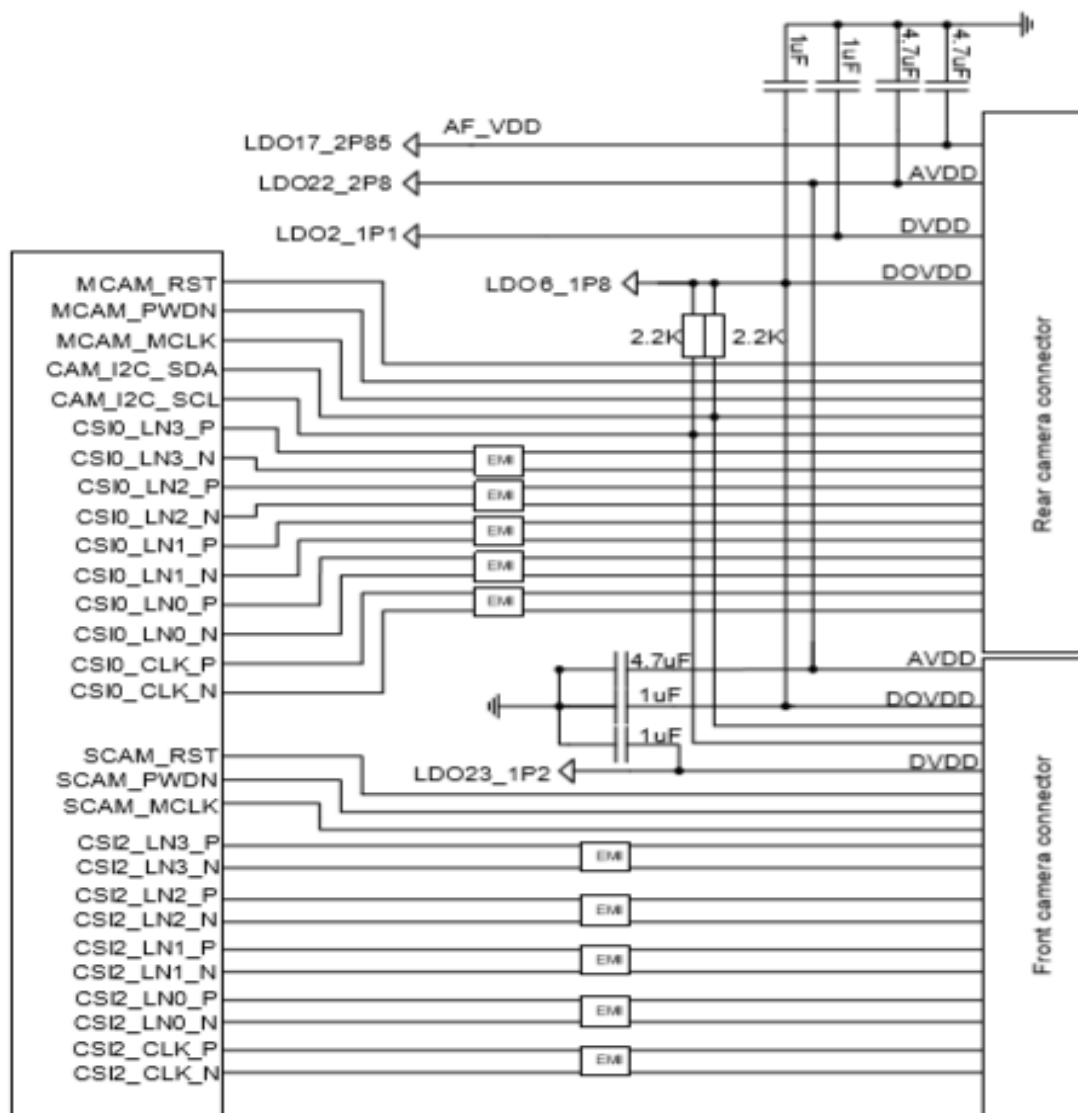
To study design and run play-store applications on 4G Mobile Trainer.

1) Camera Interfaces

4G Mobile Trainer provides two camera interfaces (4-lane+2-lane) with rear camera and front camera assembled.

- The rear camera uses a 16M pixels digital CMOS sensor with auto-focusing function.
- The front camera uses an 8M pixels CMOS sensor with auto-focusing function.

The following figure shows a reference design for camera interfaces.



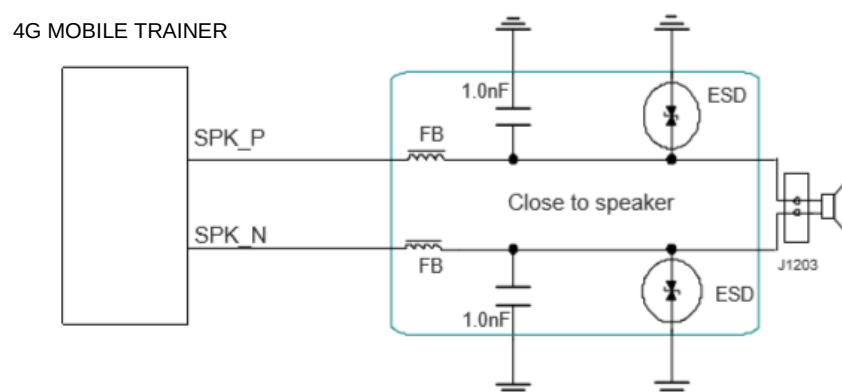


Camera Interface with Cameras Assembled

2) Audio Interfaces

4G Mobile Trainer provides three analog audio outputs including one mono loudspeaker, one mono earphone, and one stereo headset.

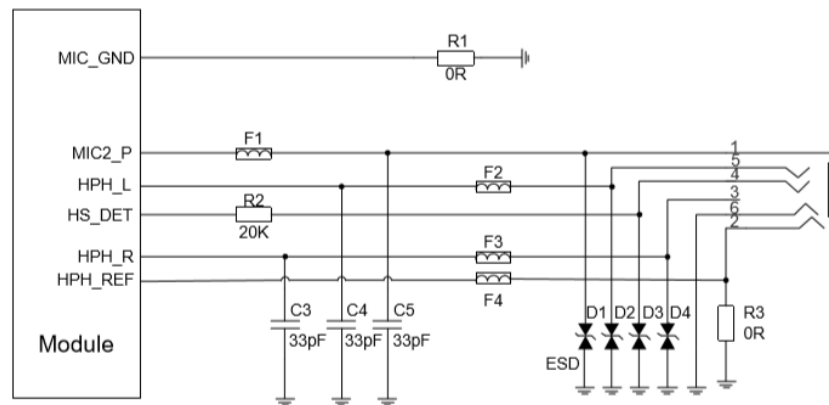
- Loudspeaker Interface
4G Mobile Trainer provides one loudspeaker interface. The following figure shows a reference circuit design for loudspeaker interface.



Reference design for Loudspeaker Interface

- Headset Interface

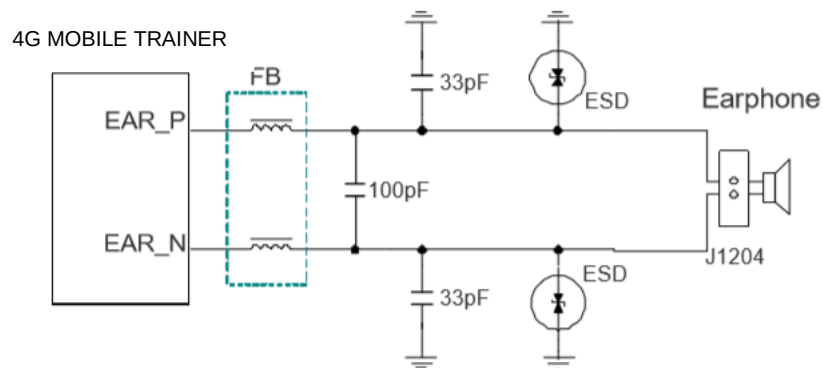
The following figure shows a reference circuit design for headset interface.



Reference design for Headset Interface

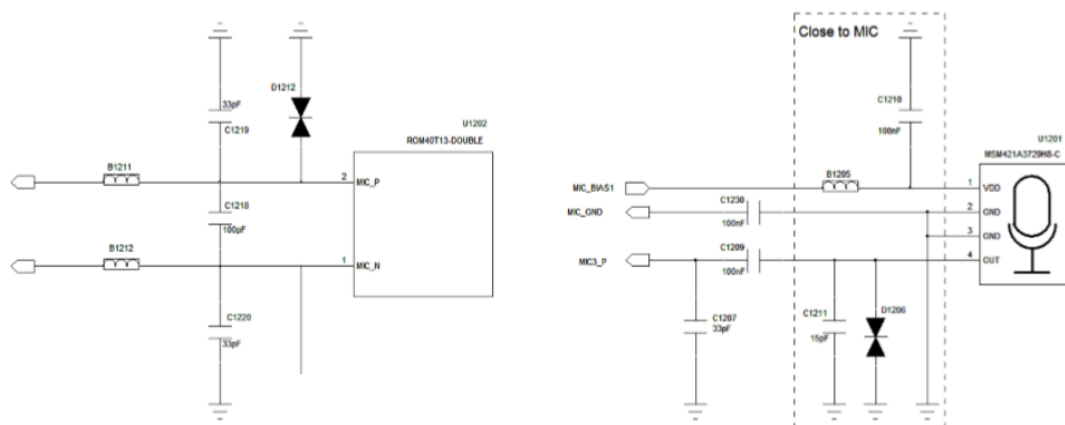
- Earphone Interface

The following figure shows a reference circuit design for earphone interface

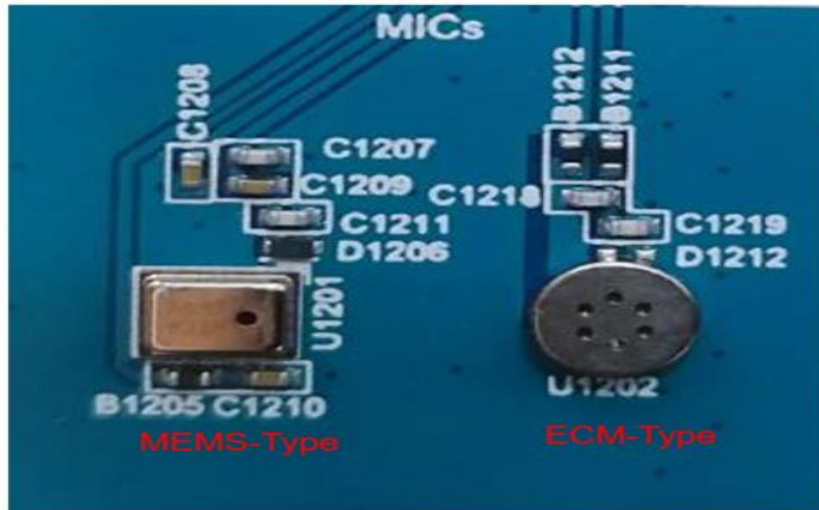


Reference design for Earphone Interface

- Microphone Interfaces



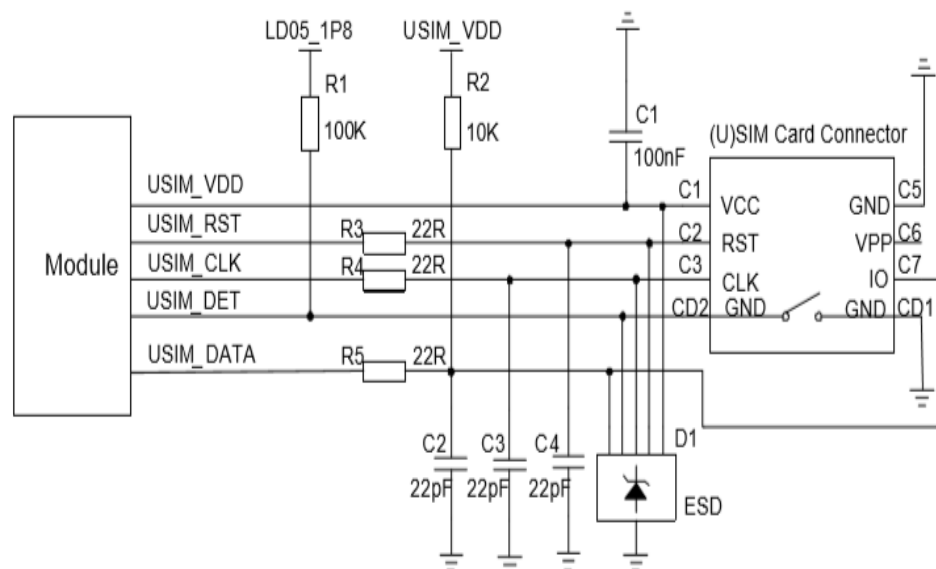
Reference design for Microphone Interfaces



MEMS-Type and ECM-Type Microphones

3) (U)SIM Interfaces:

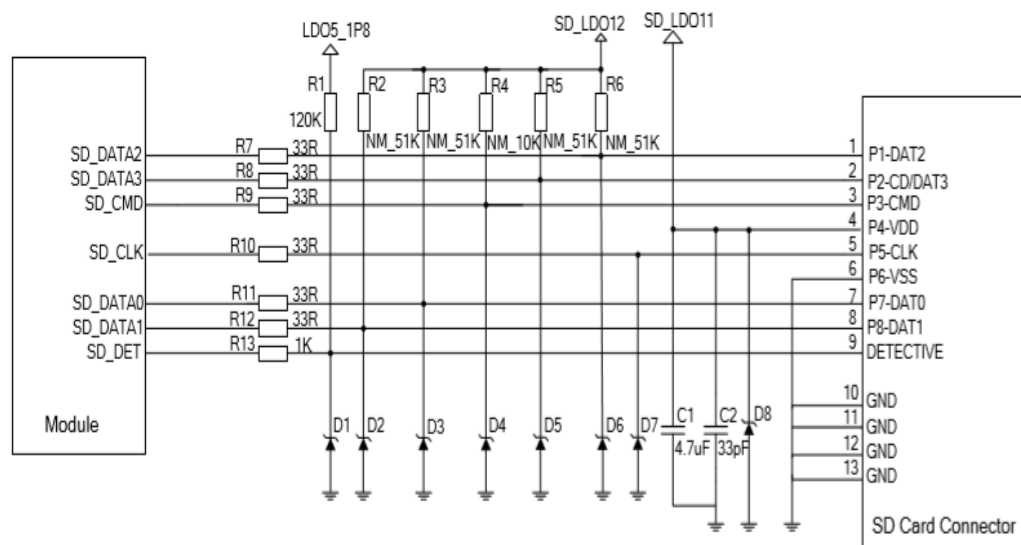
The 4G Mobile Trainer provides two 6-pin push-push type (U) SIM card (3V or 1.8V) connectors J1001 ((U) SIM1 card connector) and J1002 ((U) SIM2 card connector). The following figure shows the simplified interface schematic for J1001.



Simplified Interface Schematic for (U)SIM Card Connector

4) SD Card Interface

The following figure shows the simplified interface schematic for SD card interface on 4G Mobile Trainer.



Simplified Interface Schematic for SD Card Connector



Pin Assignment of SD Card Interface

5) Sensors

4G Mobile Trainer provides three sensors:

- I. Accelerometer and gyroscope sensor
- II. Compass Sensor
- III. Ambient light sensor and proximity sensor, as shown in the figure below.

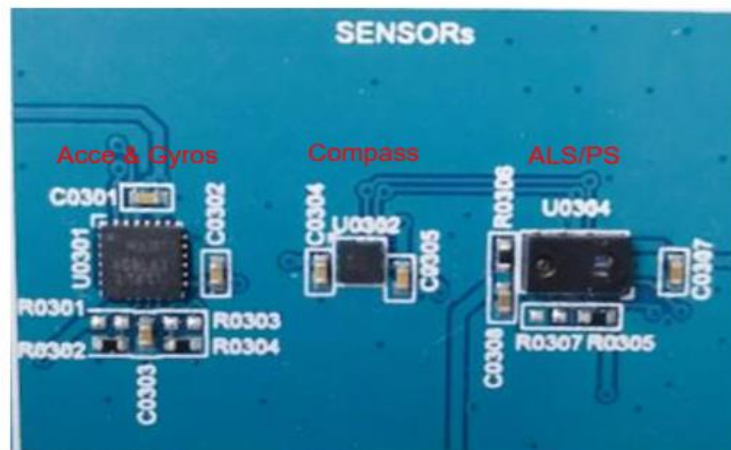
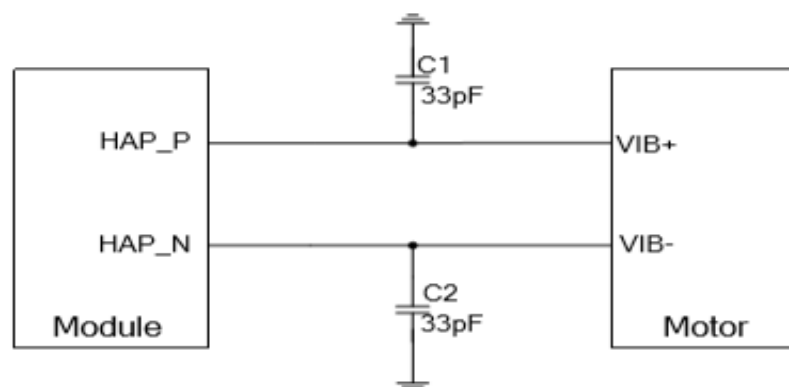


Figure 41: Sensors

Sensors

6) Vibrator

An ERM-type vibrator to test the motor driver interface of Smart modules. The following figures show a reference circuit design and the vibrator on the board.



Reference design for Vibrator

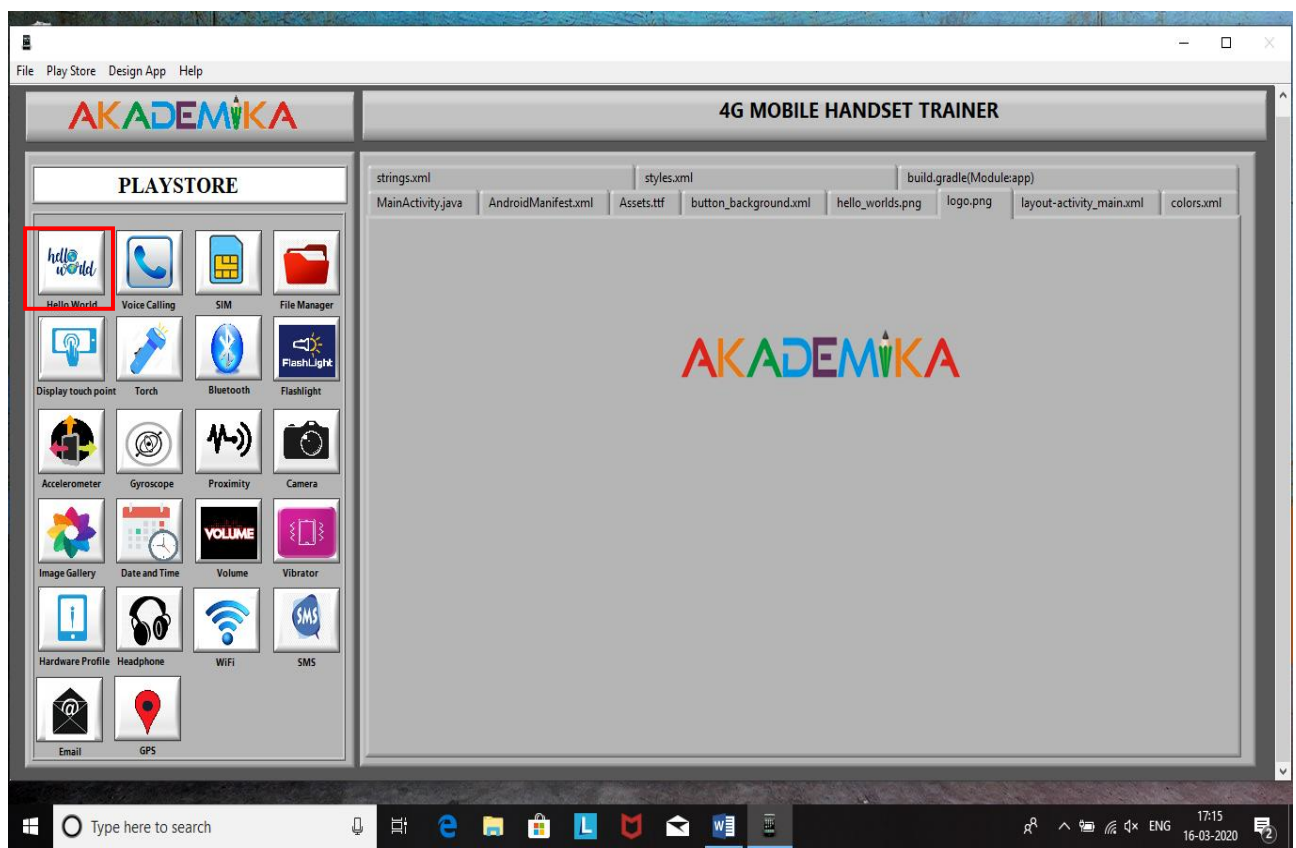


To run play-store applications on 4G Mobile Trainer follow the following procedure:

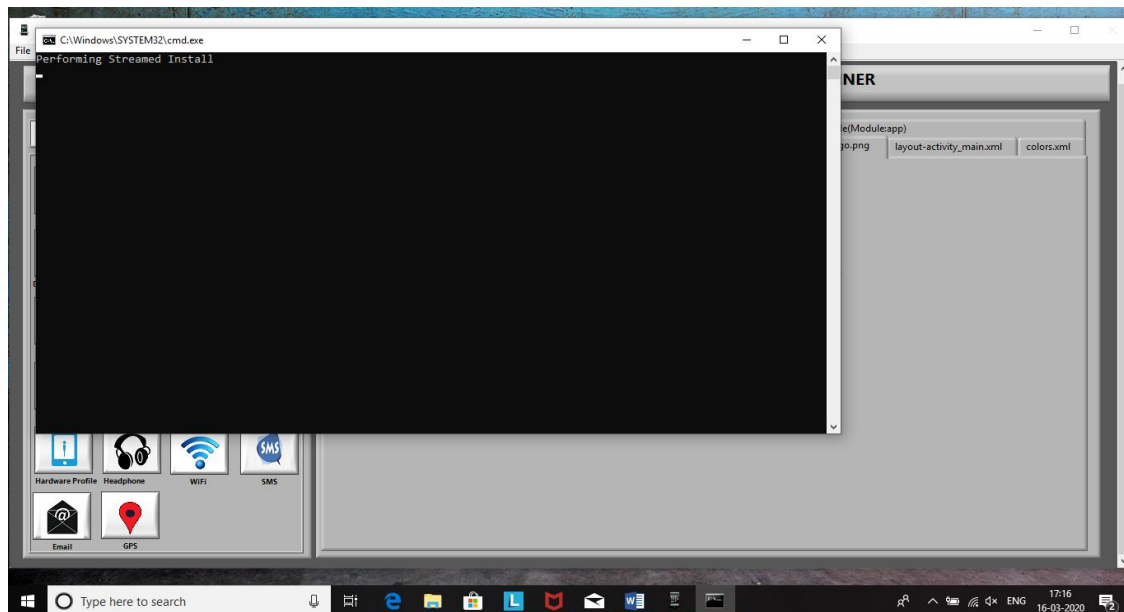
- I. Open **4G Mobile** application (Double click on following icon on desktop).



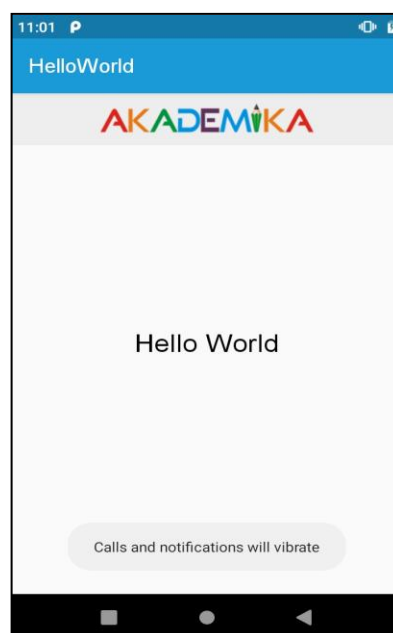
- II. Following application window will be open.



- III. Click on 'Hello World' application logo, then the command prompt window will pass the command and an application will get installed on 4G Mobile Trainer Kit.



- IV. The 'Hello World' application will be installed. Open it from the applications window on 4G Mobile trainer kit.



EXPERIMENT

STUDY OF ANDROID STUDIO PROGRAMS

STUDY OF ANDROID STUDIO PROGRAMS

To Study & execute android programs for installation of apk files on 4G mobile trainer.

ANDROID STUDIO JAVA SCRIPT FOR ACCELEROMETER APPLICATION

```
package in.akademika.com.accelerometer;

import android.graphics.Color;
import android.hardware.Sensor;
import android.hardware.SensorEvent;
import android.hardware.SensorEventListener;
import android.hardware.SensorManager;
import android.support.v7.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.view.Window;
import android.view.WindowManager;
import android.widget.TextView;
import android.widget.Toast;

import me.anwarshahriar.calligrapher.Calligrapher;

public class MainActivity extends AppCompatActivity implements
SensorEventListener {
    private SensorManager sensorManager;
    private boolean color = false;
    private View view,layout;
    private long lastUpdate;
    private TextView textView;

    /** Called when the activity is first created. */
    @Override
    public void onCreate(Bundle savedInstanceState) {
        /* requestWindowFeature(Window.FEATURE_NO_TITLE);
        getWindow().setFlags(WindowManager.LayoutParams.FLAG_FULLSCREEN,
            WindowManager.LayoutParams.FLAG_FULLSCREEN);*/

        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        Calligrapher calligrapher = new Calligrapher(this);
        calligrapher.setFont(this, "arial.ttf", true);

        view = findViewById(R.id.textView);
        layout = findViewById(R.id.layout);
        textView = findViewById(R.id.textView);
        layout.setBackgroundColor(Color.WHITE);
        sensorManager = (SensorManager) getSystemService(SENSOR_SERVICE);
        lastUpdate = System.currentTimeMillis();
    }

    @Override
    public void onSensorChanged(SensorEvent event) {
        if (event.sensor.getType() == Sensor.TYPE_ACCELEROMETER) {
            getAccelerometer(event);
        }
    }
}
```

```

    }

    private void getAccelerometer(SensorEvent event) {
        float[] values = event.values;
        // Movement
        float x = values[0];
        float y = values[1];
        float z = values[2];

        float accelerationSquareRoot = (x * x + y * y + z * z)
            / (SensorManager.GRAVITY_EARTH * SensorManager.GRAVITY_EARTH);
        long actualTime = event.timestamp;
        if (accelerationSquareRoot >= 2) //
        {
            if (actualTime - lastUpdate < 200) {
                return;
            }
            lastUpdate = actualTime;
            /*Toast.makeText(this, "Device was shuffled", Toast.LENGTH_SHORT)
                .show();*/
            if (color) {

layout.setBackgroundColor(getResources().getColor(R.color.colorAccent));
                textView.setTextColor(Color.WHITE);
            } else {
                layout.setBackgroundColor(Color.WHITE);

textView.setTextColor(getResources().getColor(R.color.colorAccent));
            }
            color = !color;
        }
    }

    @Override
    public void onAccuracyChanged(Sensor sensor, int accuracy) {

    }

    @Override
    protected void onResume() {
        super.onResume();
        // register this class as a listener for the orientation and
        // accelerometer sensors
        sensorManager.registerListener(this,
            sensorManager.getDefaultSensor(Sensor.TYPE_ACCELEROMETER),
            SensorManager.SENSOR_DELAY_NORMAL);
    }

    @Override
    protected void onPause() {
        // unregister listener
        super.onPause();
        sensorManager.unregisterListener(this);
    }
}

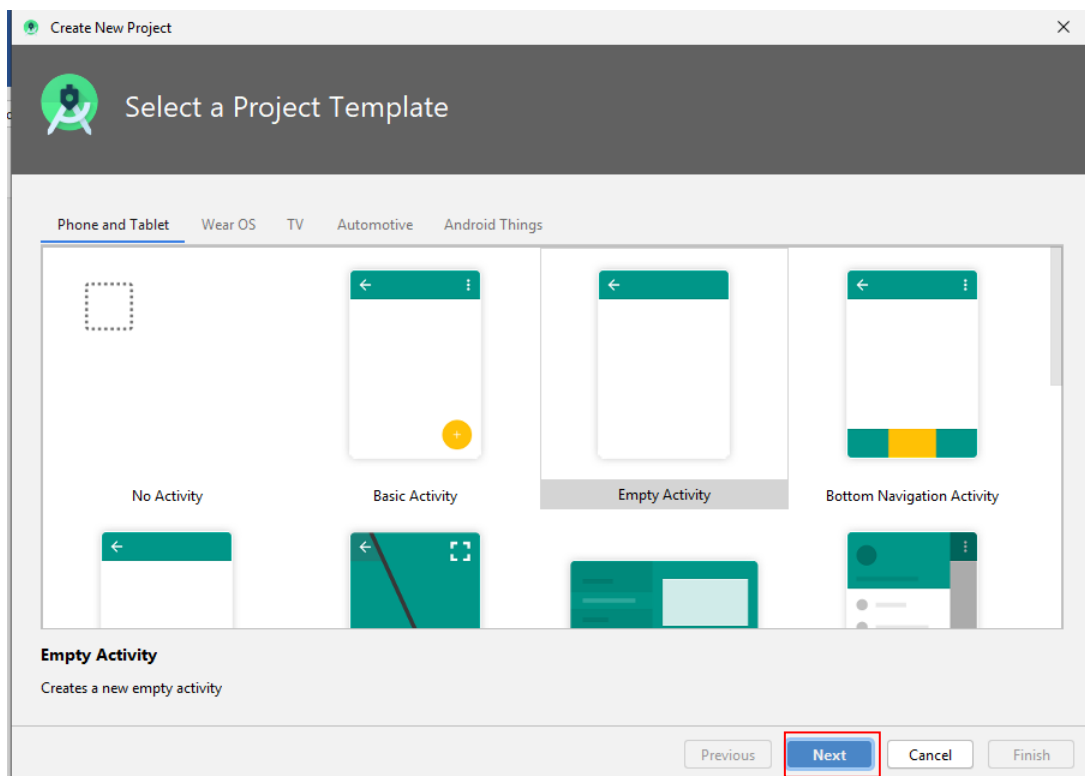
```

Follow the following procedure to development and installation of apk files on 4G mobile through Android Studio JAVA Script.

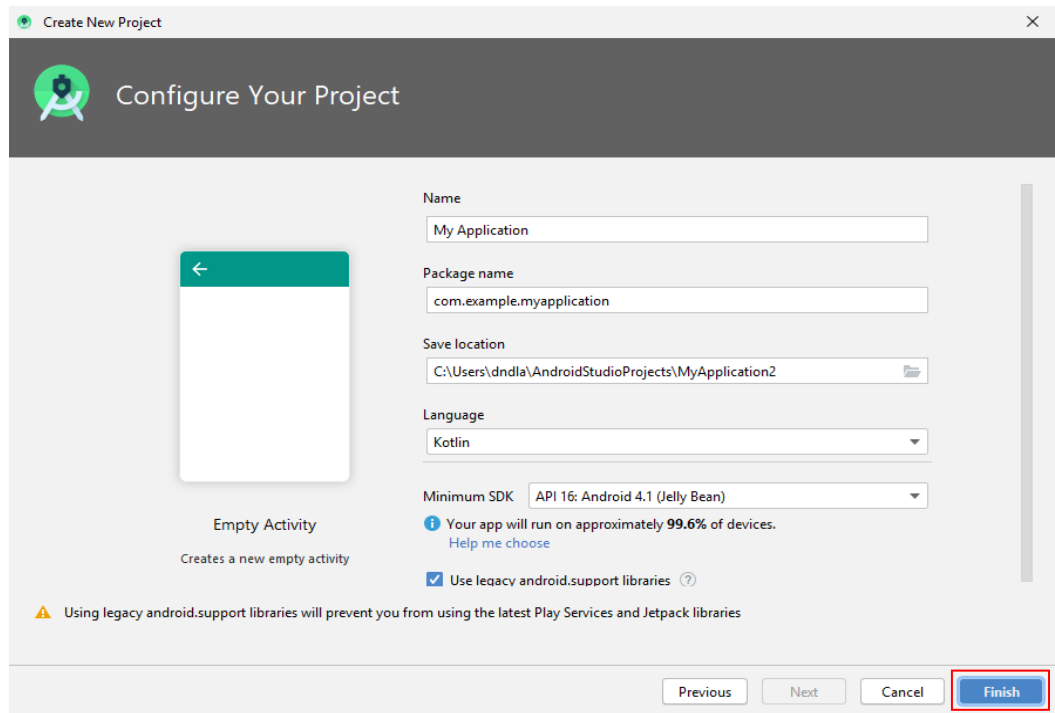
- I. Open android studio application.



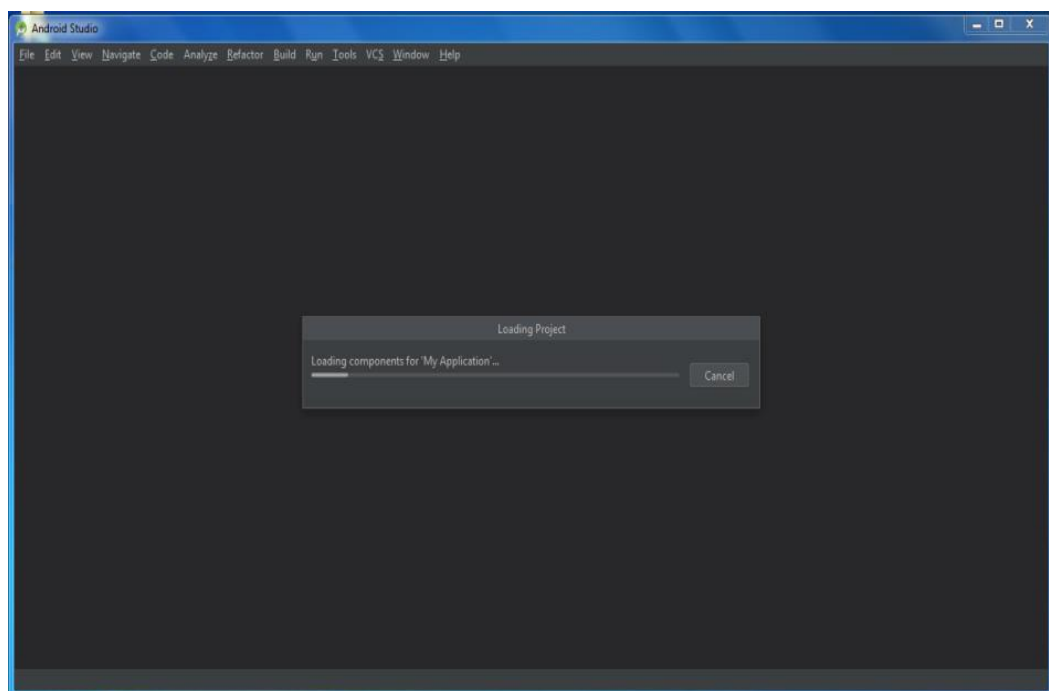
- II. Select a Project Template & click on Next.



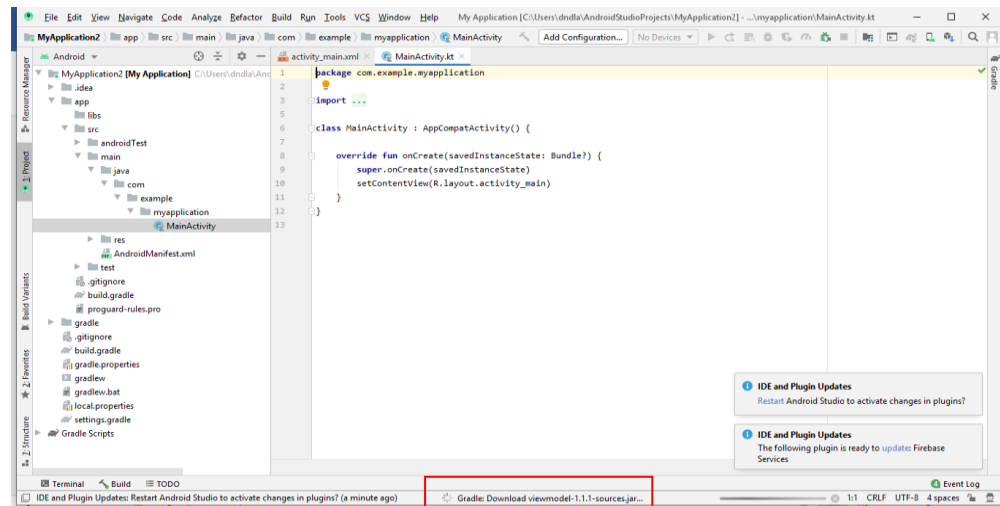
III. Configure the Project & click on Finish.



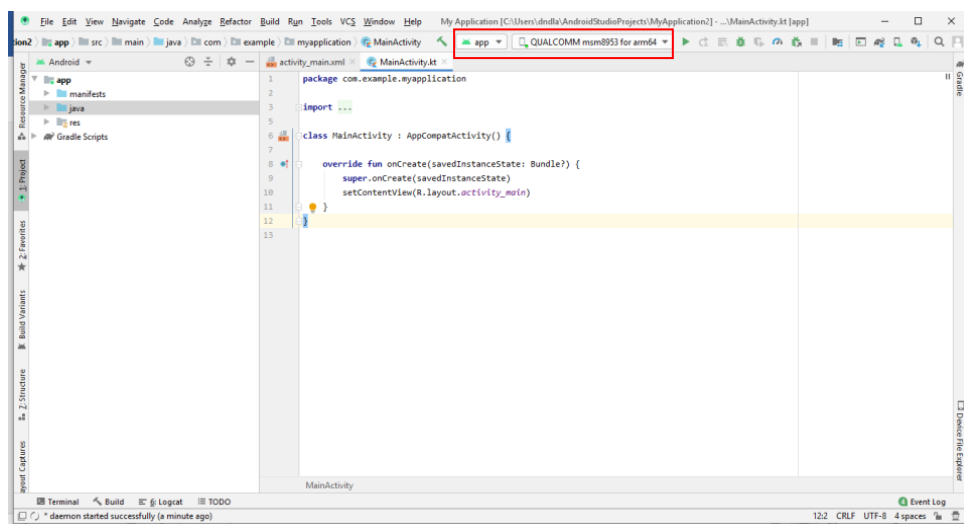
IV. Android Studio will start loading the Project.



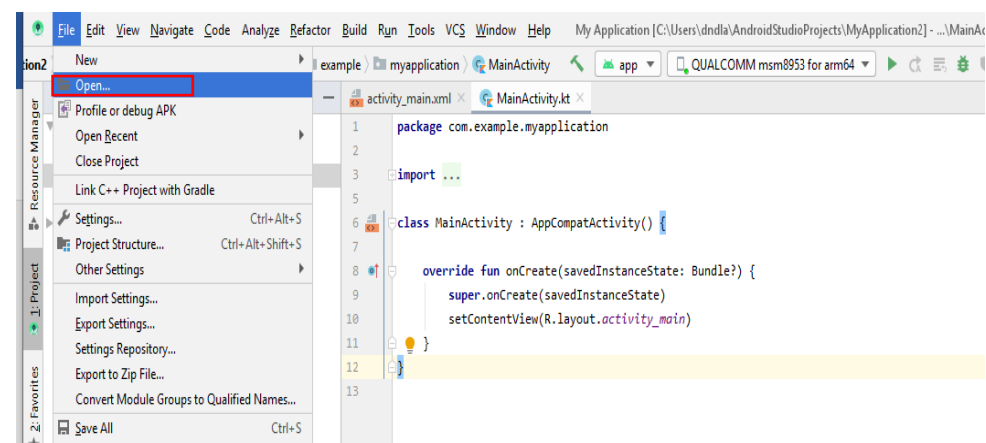
V. Wait till all Project components are downloading.



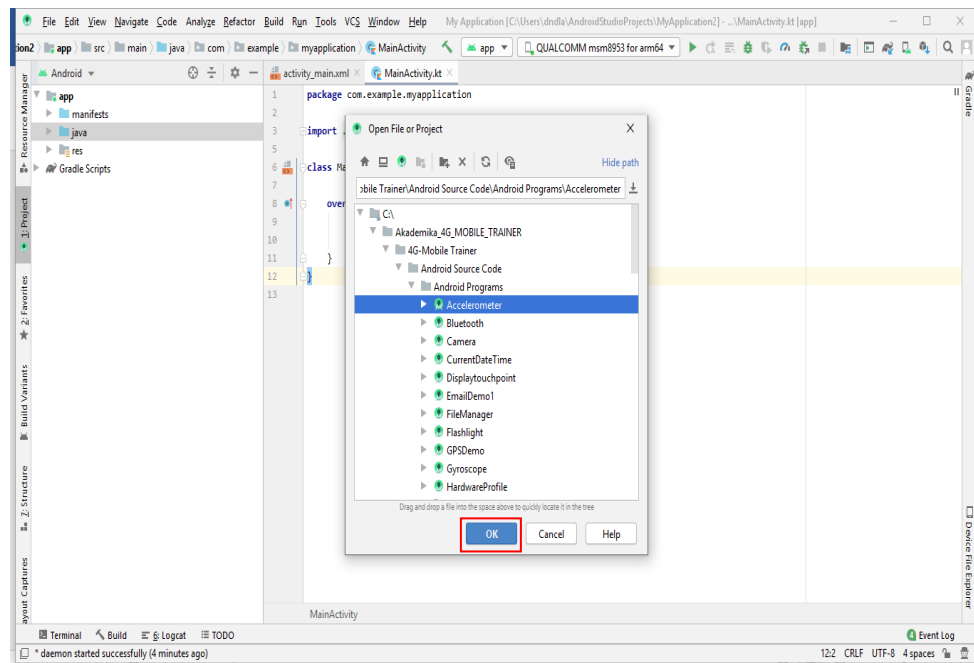
VI. Once the all the project components get downloaded then it will detect the connected device.



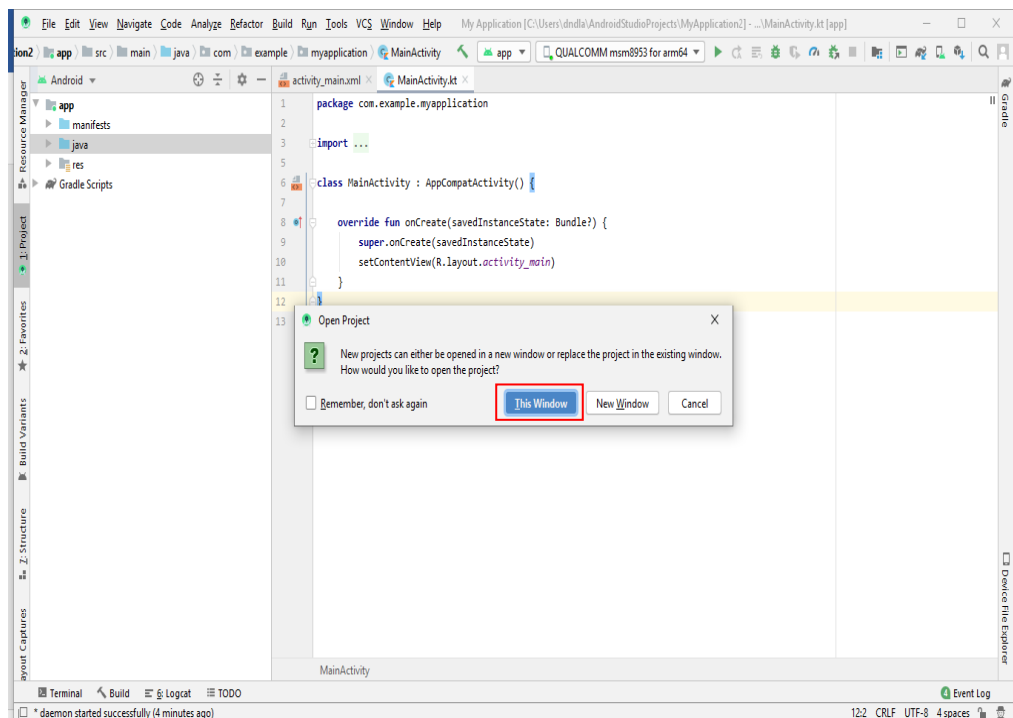
VII. Now, Go to File menu & click Open.



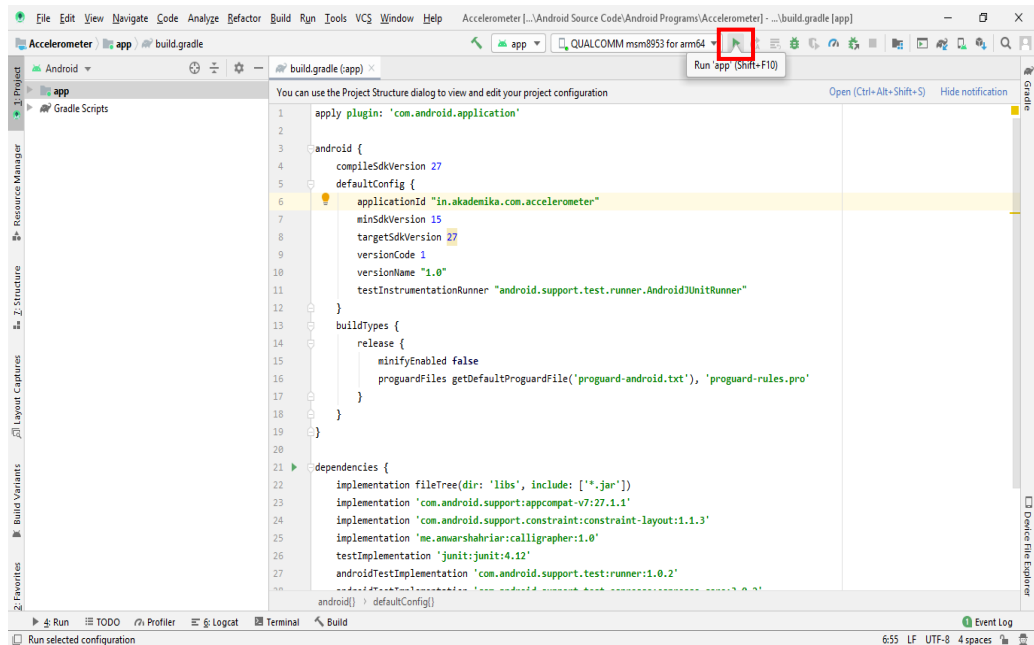
VIII. Select 4G-Mobile Trainer>>Android Source Code>>Android Programs>>Accelerometer & click OK.



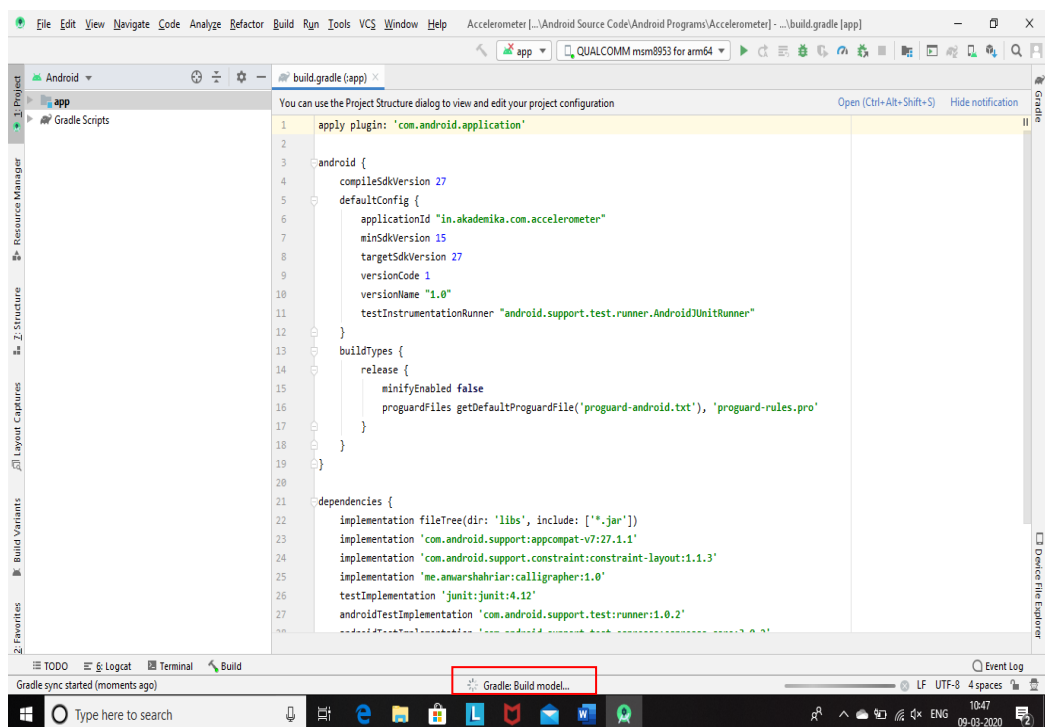
IX. Following window will be pop out, click on 'This Window'.



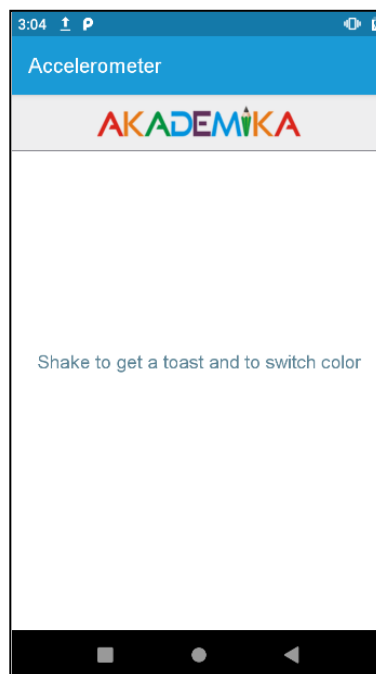
X. The program will be open, now run the program by clicking on this  button.



XI. After running the program, wait until the loading process complete.



- XII. Now apk file of the accelerometer mobile application will be sent to the 4G-Mobile Trainer Kit. Check it on Mobile Trainer Kit.



List of applications along with their android code

- Bluetooth
- Camera
- Current Date Time
- Display Touch Point
- Email Demo
- File Manager
- Flashlight
- GPS
- Gyroscope
- Headphone
- Hello World
- Hotspot Demo
- Image Gallery
- Increase Decrease Volume
- Proximity
- Sim Card
- SMS Demo
- Torch
- Vibrator
- Voice Calling
- WiFi Demo

Follow the procedure of accelerometer application, to development and installation of above listed apk files on 4G mobile through Android Studio JAVA Script.